

Standard Operating Procedures (SOP) Manual

Operations, Administration & Maintenance Handbook.

A complete guide to system health checks, user provisioning, database tasks, disaster recovery execution, and DevOps deployment playbooks.

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Restricted & Confidential

PREPARED BY

DevOps & Core Operations Group

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CHAPTER 02

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CHAPTER 03

Document Control

Revision Control, Approvers, and Distribution Directory

Summary: Details the change log, revision tracking history, and operational approval authority records for the MS Family platform.

1. Revision History

Table 1: SOP Manual Revision Log

VERSION	RELEASE DATE	SUMMARY OF CHANGES	AUTHOR	REVIEWER
v0.9.0	2026-06-15	Initial Draft of System SOP Manual.	J. Doe (DevOps)	A. Smith (Sec)
v1.0.0	2026-07-07	Enterprise Release with Local Mermaid CLI pipelines.	Core Ops Group	Board of Admins

2. Document Approvers

Table 2: SOP Operational Approvals Registry

NAME	ROLE	DEPARTMENT / TITLE	SIGN-OFF DATE	STATUS
Alice Vance	System Director	Infrastructure Operations	2026-07-07	Approved
Bob Miller	Chief Security Officer	Information Security	2026-07-07	Approved
Charles Green	Database Lead	PostgreSQL DBA	2026-07-07	Approved

3. Distribution Directory

Table 3: Document Distribution List

GROUP / ENTITY	FORMAT	ACCESS LEVEL	PURPOSE
DevOps Engineering	Digital PDF	Read / Write	Infrastructure deployment & updates
System Operations (SecOps)	Digital PDF	Read Only	Daily system health auditing
Customer Support Support	Digital PDF	Read Only	Customer triage and ticket resolutions

CHAPTER 04

Purpose, Scope & Governance

Operational Bounds, Compliance, and Business Governance Standards

1. Objectives of the Manual

This Standard Operating Procedures (SOP) Manual defines the procedures for running, monitoring, maintaining, and recovering the **MS Family** platform. It ensures operational consistency and reduces service interruptions.

2. Operational Scope

This document covers the following environments:

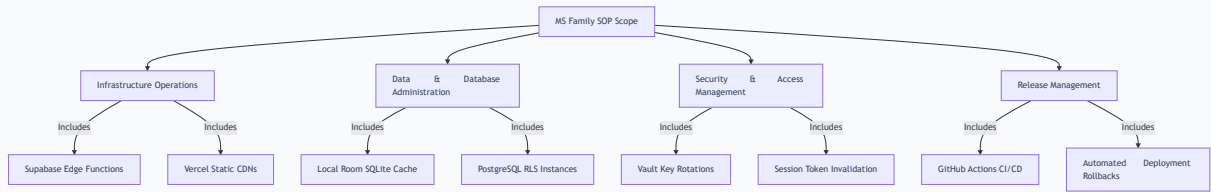


Diagram 1: SOP Scope Matrix and Lifecycle Operations



Excluded Scope Items

Hardware failures at Google Cloud (Gemini) or Supabase (PostgreSQL hosting) are managed under their respective SLAs and are excluded from this manual.

CHAPTER 05

Roles & Responsibilities RACI

Operations Staff, RACI Matrix, and System Boundaries

1. RACI Matrix

Table 4: Operational Responsibility Matrix (RACI)

OPERATIONAL TASK	SYSADMIN	DEVOPS	DEVELOPER	DBA	SECADMIN	SUPPORT	FAMILY ADMIN
System Startup & Health	A	R	C	C	I	I	I
User Provisioning	R	I	I	I	A	C	R
Database Vacuum & Index	C	C	I	R	I	I	I
Security Auditing	A	C	I	C	R	I	I
Application Deployment	A	R	R	C	C	I	I
Backup Verification	A	R	I	R	I	I	I
Disaster Recovery	A	R	C	R	R	C	I

- **R (Responsible):** The role that executes the task.
- **A (Accountable):** The role with approval authority and final ownership.
- **C (Consulted):** Roles that provide input.
- **I (Informed):** Roles notified of outcomes.

**Operational Constraint**

No single developer may hold both Accountable (A) and Responsible (R) privileges for production deployment workflows.

CHAPTER 06

Daily Operations Runbook

Daily System Health Checks and Log Auditing Procedures

SOP-OPS-01: Daily System Health Check

1. Purpose

Ensures core services (API endpoints, database pools, notification engines, and AI integrations) are operational.

2. Scope

Applies to production web apps, native bridges, and database connections.

3. Step-by-Step Procedure

- 1. API Gateway Ping:** Ping the Supabase REST endpoint: `bash curl -i -s -o /dev/null -w "%{http_code}" https://YOUR_SUPABASE_PROJECT.supabase.co/rest/v1/` *Expected Result:* HTTP Code `200` or `401` (Unauthorized indicates endpoint is live but requires credentials).
- 2. Database Connection Count:** Log into the Supabase Dashboard and check Postgres active clients.
Threshold limit: Alert if connection utilization exceeds 85% of pool bounds.
- 3. Storage Usage Check:** Run a storage directory size sweep: `sql SELECT sum(size) FROM storage.objects WHERE bucket_id = 'proofs';`
- 4. Log Sweep:** Check Sentry dashboard for errors categorized as `Fatal` or `Error`.



Exception Handling

If the REST endpoint returns HTTP 5xx, immediately route to the [Disaster Recovery SOP](#).

CHAPTER 07

User Provisioning SOP

Account Lifecycles, Onboarding, and Security Provisioning Procedures

SOP-USR-01: Member Invitation & Onboarding

1. Purpose

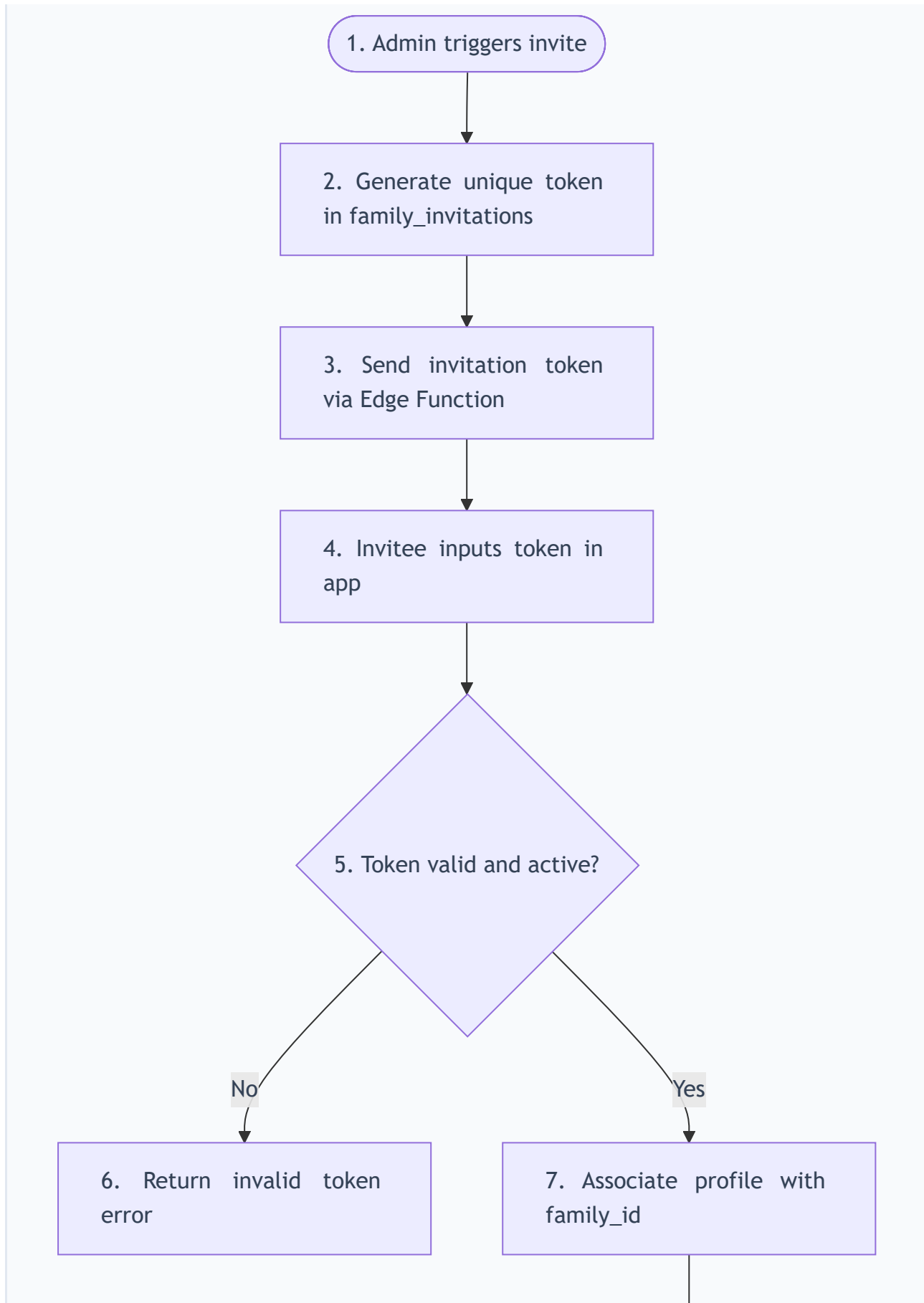
Defines the workflow for onboarding new family members.

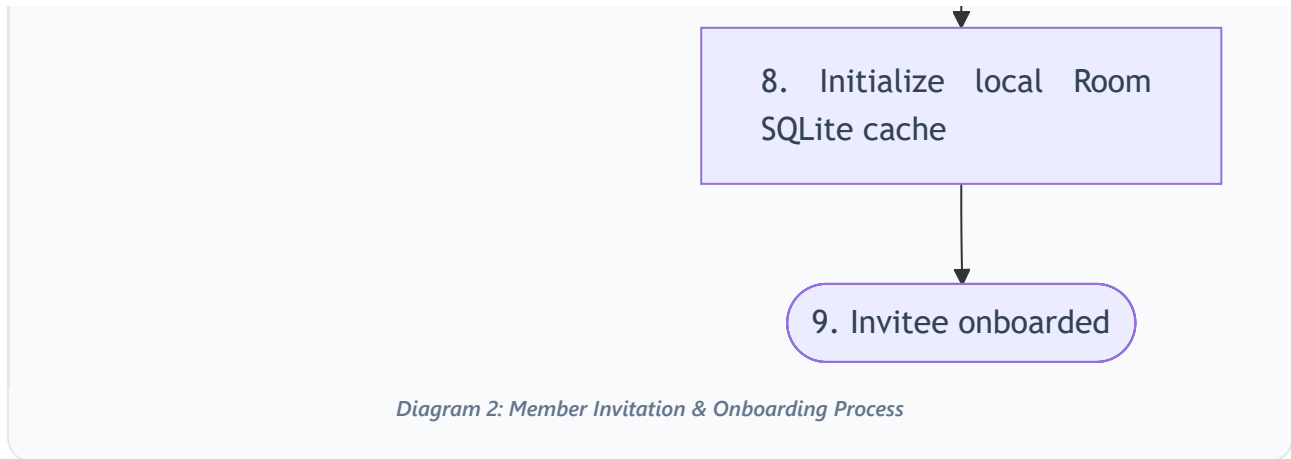
2. Prerequisites

1. Requester must hold `role = 'admin'` for the target `family_groups` container.
2. Invitee must have a registered email address.

3. Step-by-Step Procedure







4. Exception Handling

- **Expired Token:** If the token has expired, the administrator must regenerate the invite, which invalidates the old record.

CHAPTER 08

Expense & Ledger Ingestion SOP

Manual Posting, SMS Interceptions, and Monthly Close Workflows

SOP-FIN-01: Ledger Category Maintenance

1. Purpose

Maintains consistency in financial reporting by managing the categorization rules for household expenses.

2. Step-by-Step Procedure

- Deduplication Check:** When transactions are written from the Android SMS Receiver, ensure the system runs a deduplication query: `sql SELECT id FROM public.transactions WHERE sms_reference = 'SMS_UNIQUE_REF_HASH';`
- Postgres Ingestion Check:** Execute audit sweep queries once a day to check for category mismatches: `sql SELECT id, category FROM public.transactions WHERE category NOT IN ('Food', 'Utilities', 'Health', 'Travel', 'Loans', 'Other');`
- Manual Adjustments:** Correct mismatched entries by running a targeted update through the Supabase console, logging the change in the audit database.



Audit Trail Enforcement

Direct database edits are restricted to emergency scenarios and must be logged in the system audit trail.

CHAPTER 09

Bill Schedules & Overdue SOP

Bill Reminders, EMI Tracking, and Overdue Alert Workflows

SOP-BILL-01: Recurring Bill Configurations

1. Purpose

Ensures household bills and loan EMIs are tracked and reminded in advance.

2. Prerequisites

- 1. Access to the Supabase database dashboard or the system administration portal.
- 2. Active FCM keys configured for push notifications.

3. Step-by-Step Procedure

Table 5: Bill & EMI SOP Steps

STEP	ACTION	EXECUTED BY	EXPECTED OUTCOME
1	Enter bill details (due date, amount, provider)	Family Member	Scheduled entry created in database.
2	Check upcoming bills (48-hour threshold)	WorkManager Worker	Local alarm registered on client devices.
3	Dispatch alert notification	Notification Engine	User receives push alert for payment.
4	Mark bill payment status as paid	Family Member	Transaction record generated, ledger updated.

4. Exception Handling: Overdue Handling

If a bill remains unpaid past its due date: 1. The reminder frequency increases to every 24 hours. 2. A priority alert is flagged on the family dashboard interface.

CHAPTER 10

Secure Document Vault SOP

Ingestion Rules, Storage Policies, and OCR Validations

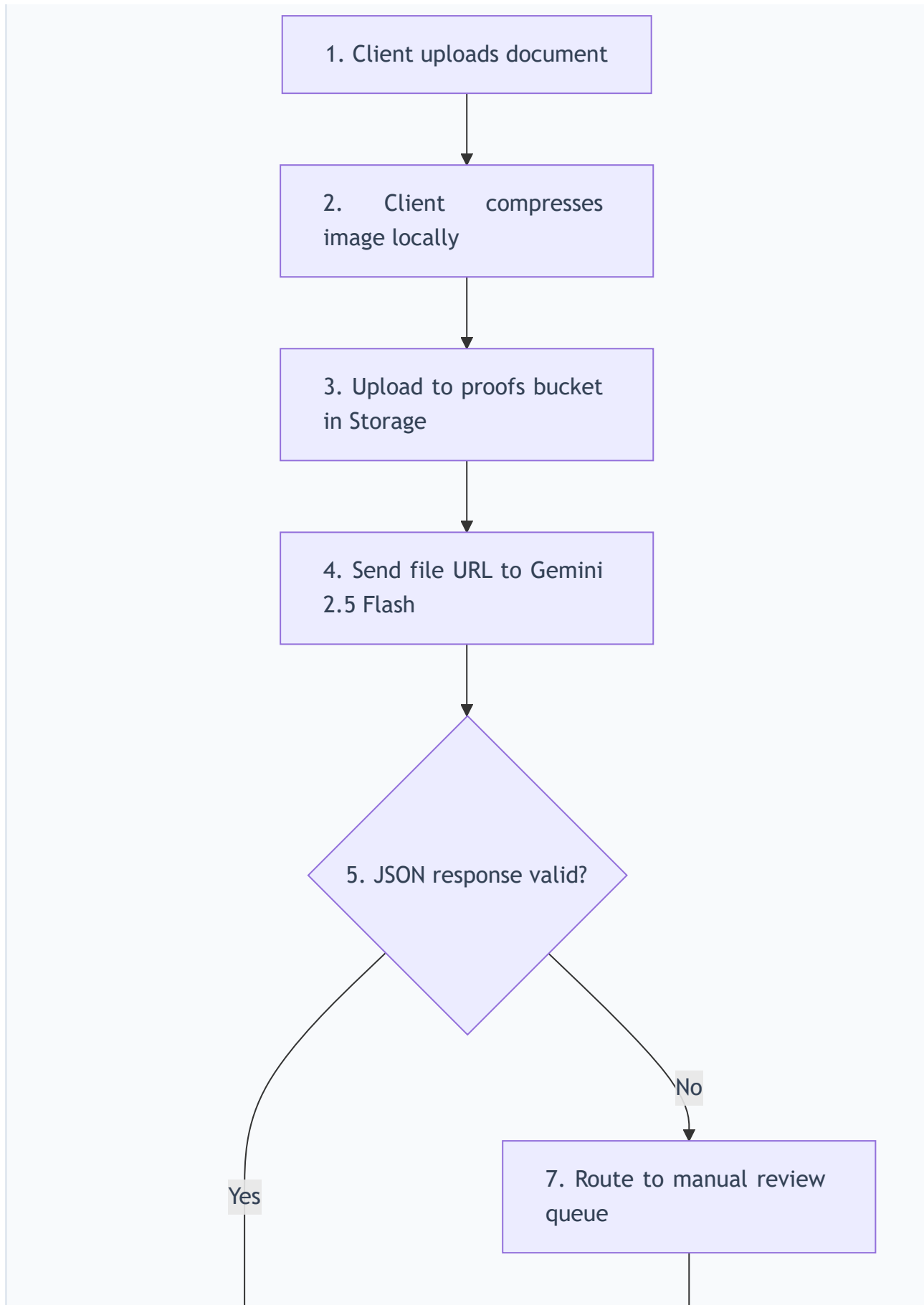
SOP-DOC-01: Ingestion & Gemini OCR Processing

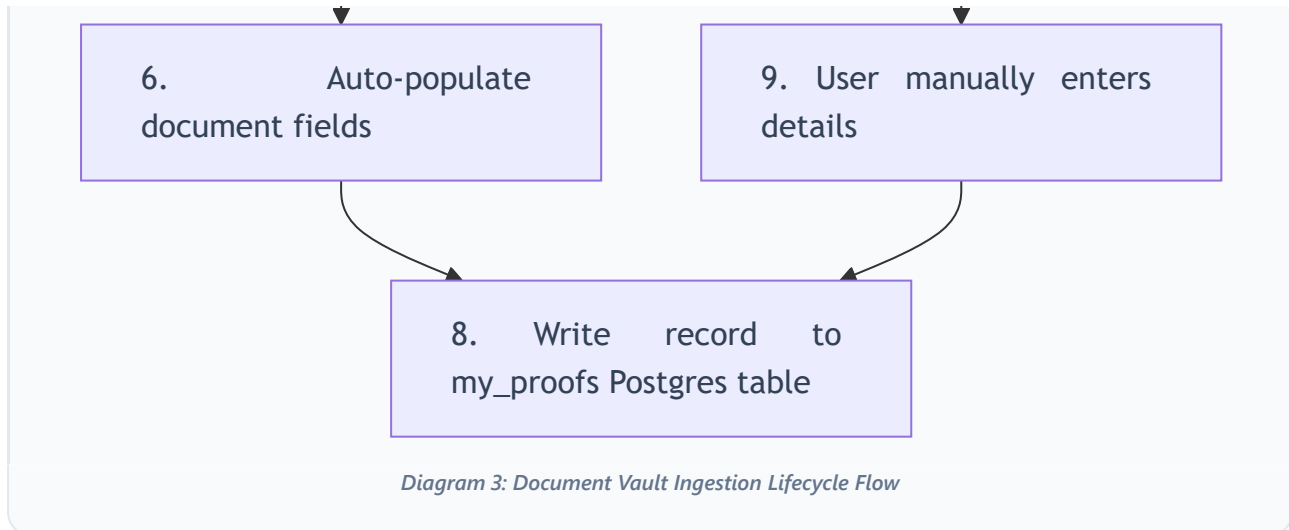
1. Purpose

SOP for uploading documents to the vault, verifying Gemini OCR extractions, and archiving files.

2. Step-by-Step Ingestion Pipeline







3. Recovery Steps: Storage Allocation Failures

If storage uploads fail due to folder allocation limits:

1. Verify user's group tier is within resource boundaries.
2. Run storage cleanup sweeps to archive historical proofs older than 24 months.

CHAPTER 11

AI Gemini API Operations SOP

Prompt Versioning, Rate Limits, and Safety Validation Policies

SOP-AI-01: Prompt Deployment & Recovery

1. Purpose

Manages changes to prompts and safety settings for the Gemini 2.5 Flash integrations.

2. Operational Boundaries

Table 6: Gemini API Rate Limits

PARAMETER	THRESHOLD	ACTION
Max Requests Per Minute	15 RPM	Throttle requests, route to local regex cache.
Safety Threshold	High Block	Return a generic validation error to the client.
Retry Limit	3 Attempts	Fail over to local text extraction.



ADR Reference: Gemini API Gateway

All Gemini API calls are routed through Supabase Edge Functions. This hides API keys and provides a layer for rate-limiting. Detailed decisions are documented in [ADR-003](#).

CHAPTER 12

Firestore Push & Reminder SOP

Notification Broker Services, Retries, and Token Expirations

SOP-NOTIF-01: FCM Token Sync & Maintenance

1. Purpose

Maintains FCM tokens to ensure push notifications are delivered reliably to mobile clients.

2. Step-by-Step Procedure

- Token Refresh:** When a user launches the app, compare the client device token with the `fcmtokens` record:

```
sql SELECT token FROM public.fcm_tokens WHERE user_id = 'USER_UUID' AND device_id = 'DEVICE_UUID';
```
- Invalid Token Cleanup:** Remove inactive tokens (e.g., those returning `UNREGISTERED` errors from FCM API calls):

```
sql DELETE FROM public.fcm_tokens WHERE last_used < now() - INTERVAL '60 days';
```
- Queue Sweeper:** Run cron checks to identify and retry failed notification queue tasks.

CHAPTER 13

Database & Storage Backup SOP

Backup Schedules, pg_dump configurations, and Restore Validations

SOP-BCK-01: Automated Database Backup

1. Purpose

Protects against data loss by maintaining regular backups of the PostgreSQL database and storage configurations.

2. Backup Schedule

Table 7: System Backup Schedules

TARGET	TYPE	FREQUENCY	LOCATION	RETENTION
Postgres Database	Logical pg_dump	Daily at 01:00 UTC	AWS S3 Bucket	30 Days
Postgres Database	Physical WAL	Continuous	AWS S3 Bucket	7 Days
Storage (Proofs)	Sync Copy	Daily at 02:00 UTC	Secondary GCS Bucket	90 Days

3. Step-by-Step Restore Procedure

- 1. **Provision Test DB Instance:** Launch a clean PostgreSQL server in staging.
- 2. **Fetch Backup:** Download the target backup file from the S3 storage bucket.
- 3. **Execute Restore:** `bash pg_restore --no-owner --clean -h localhost -U postgres -d staging_db backup_file.dump`
- 4. **Data Verification:** Query row counts for verification.

CHAPTER 14

Disaster Recovery Playbook

Service Recovery Runbooks, Failures, and RTO/RPO Metrics

1. Operational RTO & RPO Targets

- Recovery Time Objective (RTO): < 4 Hours (Maximum allowable downtime).
- Recovery Point Objective (RPO): < 24 Hours (Maximum allowable data loss window).

2. Server Failure Recovery Procedure

In the event of a total Vercel edge CDN or static assets server failure:

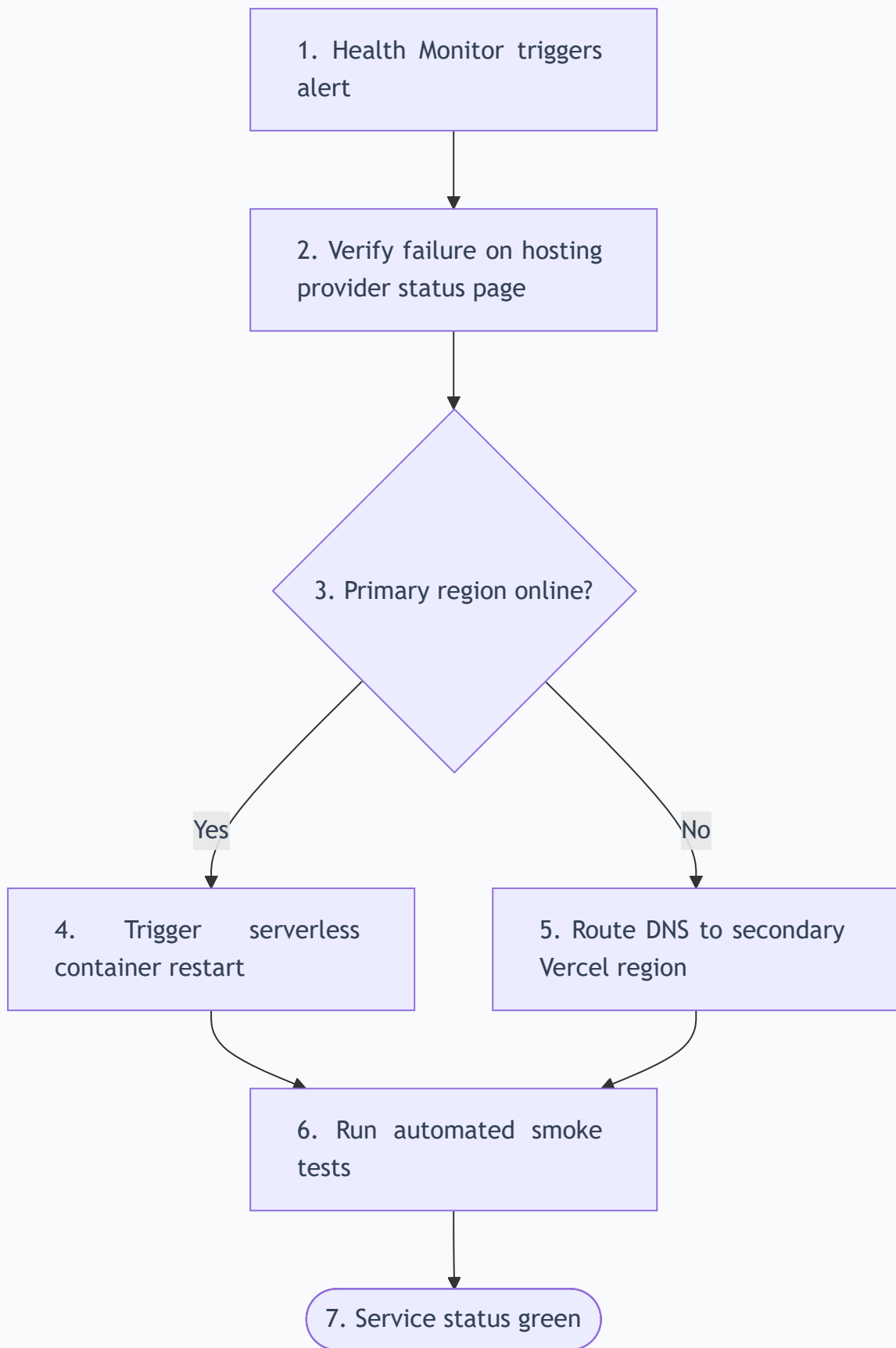


Diagram 4: Infrastructure Recovery Process

**DR Action Security Authorization**

DNS failover modifications require two-factor authorization approval from the CSO or designated Operations Lead.

CHAPTER 15

Encryption, Keys & Audits SOP

Secrets Rotation, Key Management, and OWASP Auditing Guidelines

SOP-SEC-01: API Secrets & Key Rotation

1. Purpose

Protects systems by regularly rotating API keys, database access tokens, and encryption keys.

2. Rotation Schedule

Table 8: Cryptographic Key Rotations

KEY TYPE	IDENTIFIER	FREQUENCY	ACTIONS
Supabase Anon Key	<code>apikey</code>	180 Days	Regenerate in dashboard, update Vercel environment.
Google Gemini Key	<code>VITE_GEMINI_API_KEY</code>	90 Days	Generate in Google AI Studio, update Edge variables.
JWT Signing Secret	<code>JWT_SECRET</code>	365 Days	Regenerate in Auth settings, invalidating current sessions.



Secrets Safety Policy

API keys must not be hardcoded in the codebase. Always load secrets from environment variables or a secure key management service.

CHAPTER 16

CI/CD Release & Rollback SOP

Promotion Pathways, Production Releases, and Rollback Procedures

SOP-DEP-01: Production Code Deployment

1. Purpose

Defines the process for releasing software updates to production.

2. Prerequisites

1. All linting and TypeScript checks must pass in GitHub Actions.
2. QA lead must sign off on the release build.

3. Step-by-Step Procedure

1. **Staging Deploy:** Push the release branch to `staging`. Trigger integration tests.
2. **Production Deploy:** Merge staging branch into `main`. GitHub Actions builds and pushes the static assets to Vercel production: `bash vercel --prod --token=VERCEL_PROD_DEPLOYMENT_TOKEN`
3. **Smoke Tests:** Execute post-deploy smoke tests.

4. Rollback Procedure

If smoke tests fail post-deployment: 1. Locate the last successful deployment commit in GitHub. 2. Revert the production deployment via Vercel: `bash vercel rollback DEPLOYMENT_ID --token=VERCEL_PROD_DEPLOYMENT_TOKEN`

CHAPTER 17

Health Monitoring & Escalations

Operational Dashboards, Sentry Alert Triggers, and Triage Escalations

SOP-MON-01: Performance Alert Thresholds

1. Purpose

Identifies performance degradation before it impacts users.

2. Monitoring Targets

Table 9: Service Alert Thresholds

TARGET METRIC	NORMAL BOUNDS	ALERT THRESHOLD	ACTION
API Latency	\$ < 180 ext{ ms}\$	\$ > 350 ext{ ms}\$	Log warning, profile slow queries.
DB CPU Load	\$ < 35\%\$	\$ > 75\%\$ for 5m	Ping DB Administrator to check query locks.
Sentry Error Rate	0 Per Min	\$ > 5\$ Errors / Min	Notify DevOps On-Call Team via PagerDuty.



Sentry Integration

Frontend runtime exceptions are forwarded to Sentry, grouping duplicate events to prevent alert fatigue.

CHAPTER 18

Incident Severity SOP

Severity Classification levels, Timelines, and Postmortems

SOP-INC-01: Incident Severity Matrix

1. Purpose

Defines response times and triage priorities during system incidents.

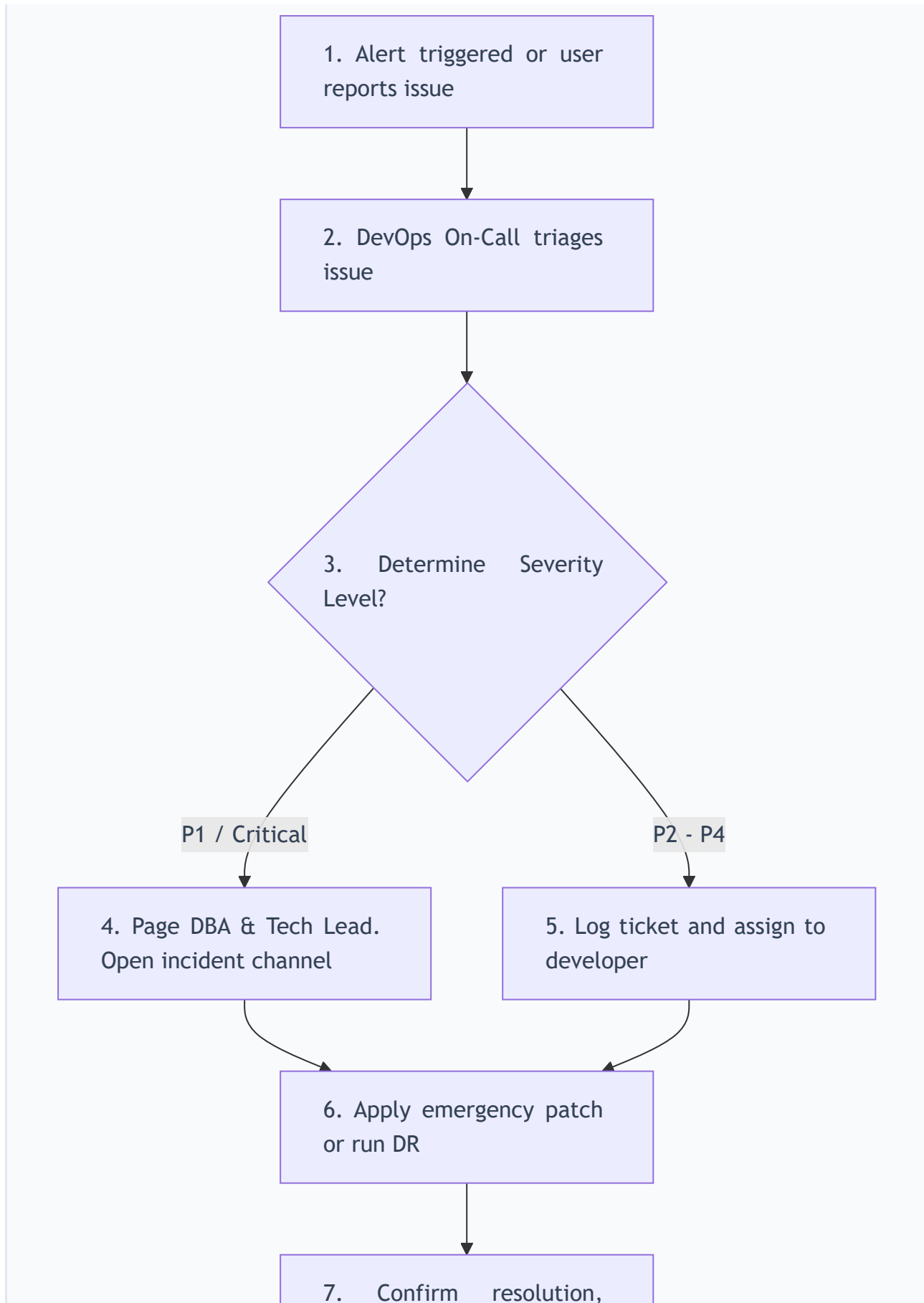
2. Severity Classification Matrix

Table 10: Incident Severity Levels

SEVERITY	DESCRIPTION	INITIAL RESPONSE	RESOLUTION SLA
P1 - Critical	Core platform down (login fails, database unreachable).	< 15\$ Minutes	< 2\$ Hours
P2 - Major	Single module degraded (OCR failing, WebRTC voice cuts).	< 30\$ Minutes	< 8\$ Hours
P3 - Minor	Minor functionality issue (dashboard graph rendering lag).	< 2\$ Hours	< 24\$ Hours
P4 - Low	Cosmetical issue or UI styling bugs.	< 24\$ Hours	Next Sprint

3. Step-by-Step Incident Triage





close ticket



8. Schedule postmortem
within 48 hours

Diagram 5: Incident Management Process Flow

CHAPTER 19

Log Rotations & DB Vacuum SOP

Regular Database Maintenance and Log Rotation Runbooks

SOP-MNT-01: PostgreSQL Database Maintenance

1. Purpose

Improves database query performance and manages storage space through regular database maintenance.

2. Step-by-Step Database Maintenance Runbook

Table 11: Database Maintenance Actions

STEP	ACTION	COMMAND	FREQUENCY
1	Run VACUUM to reclaim space	<code>VACUUM (ANALYZE, VERBOSE) public.transactions;</code>	Weekly
2	Rebuild indexes to improve queries	<code>REINDEX TABLE public.transactions;</code>	Monthly
3	Clean up notification history	<code>DELETE FROM public.notifications WHERE created_at < now() - INTERVAL '90 days';</code>	Monthly
4	Archive historical sync logs	<code>COPY public.sync_logs TO PROGRAM 'gzip > sync_logs_archive.gz';</code>	Quarterly

3. Recovery Steps: Query Lockups

If maintenance commands hang due to database lockups: 1. Find the blocking process PID: `sql SELECT pid, query, state FROM pg_stat_activity WHERE state != 'idle';` 2. Terminate the blocking query: `sql SELECT pg_terminate_backend(BLOCKING_PID);`

CHAPTER 20

Runbook & Troubleshooting Index

Operational Runbooks for Database Errors and API Connection Failures

Operational Runbook Directory

Runbook 01: Database Connection Failures

- **Symptoms:** App displays `Database Connection Timeout` or returns HTTP `504` Gateway Timeout.
 - **Possible Causes:** Database connection pool is exhausted, or the Supabase instance is down.
 - **Resolution Steps:**
 - Ping the database using PGAdmin or psql: `bash psql -h db.YOUR_PROJECT.supabase.co -U postgres -d postgres -c "SELECT 1;"`
 - If the connection fails, check Supabase service status.
 - If the connection succeeds but latency is high, check connection pool allocation: `sql SELECT count(*), state FROM pg_stat_activity GROUP BY state;`
 - Terminate idle connections to free up slots in the pool.
-

Runbook 02: WebRTC Audio Connection Failures

- **Symptoms:** VoIP calls disconnect immediately, or users experience silent calls.
- **Possible Causes:** NAT firewall is blocking connection paths, or ice candidates are failing.
- **Resolution Steps:**
 - Verify STUN/TURN server configuration inside `CallContext.tsx`.
 - Check ICE candidate exchange logs in WebRTC inspector.
 - Fail back to public Google STUN server (`stun:stun.l.google.com:19302`) if necessary.

CHAPTER 21

Maintenance & Release Checklists

Operational checklists for System Maintenance and Code Releases

1. Daily Operations Checklist

- ☐ Confirm daily pg_dump backups completed successfully at 01:00 UTC.
 - ☐ Verify Supabase database connection pool utilization is $< 80\%$.
 - ☐ Review Sentry dashboard for new critical errors or exceptions.
 - ☐ Check Gemini API rate limit warnings and usage costs in Google Cloud Console.
-

2. Production Code Release Checklist

- ☐ Verify TypeScript compile and build tests pass in GitHub Actions.
- ☐ Run schema migrations on staging database and verify compatibility.
- ☐ Execute unit test coverage checks (coverage target $\geq 80\%$).
- ☐ Confirm Vercel production preview builds look correct.
- ☐ Obtain release authorization from the Operations Lead.
- ☐ Execute deploy and run post-deploy smoke tests.

CHAPTER 22

Operational Report Templates

System Administration Templates for Incidents and Change Requests

Template: Incident Postmortem Report

=====

INCIDENT POSTMORTEM REPORT

=====

Incident ID: INC-2026-[NUMBER]

Severity: [P1 / P2 / P3]

Subject: [Incident Summary]

Date of Issue: YYYY-MM-DD

On-Call Lead: [Name]

1. INCIDENT TIMELINE

- HH:MM UTC : Alert triggered in health monitor.

- HH:MM UTC : Triage begun by On-Call engineer.

- HH:MM UTC : Issue isolated to database query lockup.

- HH:MM UTC : DB connections cleared, service restored.

2. ROOT CAUSE ANALYSIS

Explain the root cause of the issue:

[Insert detailed DBA / Sentry log analyses here]

3. CORRECTIVE ACTIONS

List the actions required to prevent recurrence:

- [] Task 1: Add indexes to slow queries.

- [] Task 2: Update alert threshold configs.

=====

CHAPTER 23

Appendix & Abbreviations

Glossary, Emergency Contacts, and Version History

1. Operational Abbreviations

- **SOP:** Standard Operating Procedures
- **RTO:** Recovery Time Objective
- **RPO:** Recovery Point Objective
- **DBA:** Database Administrator
- **RLS:** Row-Level Security
- **FCM:** Firebase Cloud Messaging
- **VOIP:** Voice over IP (WebRTC calls)

2. Emergency Contact Directory

Table 12: Operations Emergency Directory

CONTACT GROUP	ROLE	CHANNEL / PHONE	PRIORITY
DevOps On-Call	Infrastructure Support	PagerDuty / On-Call Channel	P1 / Urgent
Database Lead	DBA Escapes	extension 4402	P1 / Critical
CSO Office	Security Breach Reporting	extension 9110	Urgent